

GEOLOGICALLY RESOLVING DIPPING DEPOSITS OF YAMPA BASIN (G.R.I.D.D.Y.): NEW INTERPRETATIONS OF YAMPA RIVER BASIN EVOLUTION FROM REFLECTION SEISMIC SOUNDING

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ABSTRACT

Steamboat Springs, Colorado lies within a geologically diverse setting characterized by hydrothermal activity, subsurface faulting, glacial deposits, and active fluvial erosion, yet the area remains largely uncharacterized geophysically at depth. In this paper, we present a reflection seismic characterization of the Yampa River Basin south of Steamboat Springs using 12 km of vibroseis data acquired along Routt County Roads (RCR) 14 and 22, integrated with an existing 8 km profile along Colorado Highway (HWY) 131. Depth-migrated sections were interpreted in Petrel to produce a 3D structural model of the subsurface. We initially hypothesized a four-layer subsurface model composed of a Precambrian crystalline basement, a depositional package of the Browns Park Formation, a glacial gravel unit, and a surficial alluvial unit. However, 3D seismic interpretation revealed a five-layer stratigraphic model of the area, with the Browns Park Formation resolved as two distinct packages, a younger unit overlying an older unit. Normal faulting post-dating Browns Park deposition and pre-dating glaciation is interpreted along RCR-22 and between RCR-14 and HWY 131, with an additional fault inferred near RCR-14 from abrupt lateral changes in subsurface geometry and the structurally controlled trajectory of the Yampa River. These results provide the first geophysical constraints on subsurface fault geometry in this area, with implications for structural controls on regional hydrothermal systems and seismic hazard potential.

1. INTRODUCTION

The Yampa River Basin of northwestern Colorado sits at the intersection of several active geological processes such as hydrothermal activity, subsurface faulting, glacial deposits, and active fluvial erosion, yet the subsurface architecture governing these systems remains poorly constrained. Existing fault interpretations in the region rely predominantly on surface mapping and regional structural models, leaving the depth extent, geometry, and displacement history of basin-bounding and intrabasin faults largely unresolved (Madole, 1991). This is a critical limitation: fault geometry at depth exerts fundamental control on fluid pathways in hydrothermal systems,

on crustal mechanical behavior during seismic events, and on the stratigraphic architecture of basin fill. Without subsurface constraints, these relationships cannot be meaningfully evaluated in the Yampa Basin.

Precambrian crystalline basement underlies the basin and is expected to produce a strong seismic reflection at depth due to the sharp impedance contrast with overlying sedimentary sequences, allowing us to constrain basin geometry and thickness directly from our data. The Browns Park Formation forms the primary basin fill above the basement and has been mapped extensively at the surface, but its subsurface geometry and relationship to intrabasin faulting remain poorly documented. Pleistocene glaciation subsequently carved valley systems and deposited outwash across the upper basin, producing a thick sequence of glacial and alluvial deposits that overlie the Browns Park Formation and continue to influence modern drainage patterns (Leonard et al., 2017). Together, these stratigraphic relationships make the Yampa Basin well-suited to reflection seismic imaging, which can resolve the depth and geometry of each of these units and the structural boundaries between them.

In this paper, we present an integrated geophysical characterization of the Yampa Basin south of Steamboat Springs using vibroseis reflection seismic data acquired along RCR 14 and 22, integrated with existing profiles along HWY 131. We use these data to constrain basin geometry, stratigraphic layering, and subsurface fault structures that have not previously been mapped in this area, with implications for understanding the structural controls on the region's hydrothermal systems and seismic hazard potential.

2. METHODS

2.1. Data Acquisition

In 2026, two seismic reflection surveys were conducted along RCR-14 (north-south) and RCR-22 (east-west). RCR-22 was selected because it intersects both RCR-14 and HWY 131, the latter surveyed in 2022, providing two approximately parallel north-south lines connected by an orthogonal transect. This geometry enables a 3D structural interpretation from three 2D survey lines. Along RCR-14, 787 GTI NRU N1 single-

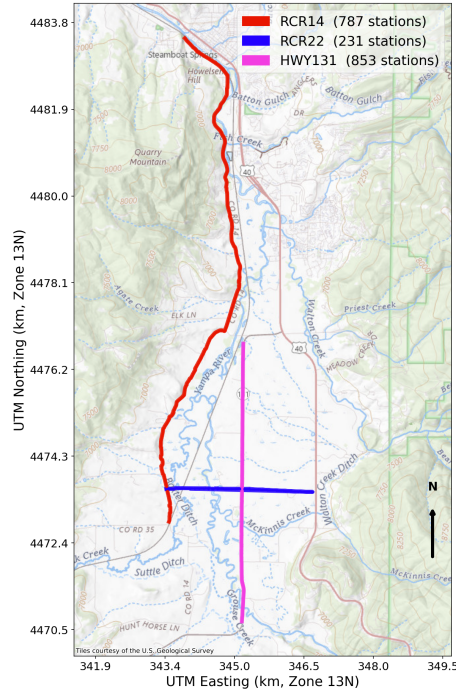


Fig. 1. Layout of Seismic Surveys in the Yampa River Basin

component seismic nodes were deployed at 15 m spacing over 11 km; 209 nodes were deployed at the same spacing along the western side of RCR-22 over 3 km.

2.2. Seismic Processing

Shallenberger et al. developed a comprehensive seismic processing workflow using Halliburton SeisSpace to optimize the signal-to-noise ratio and resolve near-surface anomalies for the Yampa River Basin dataset. Following dataset merging and datum statics correction, noise attenuation was achieved through frequency-wavenumber (f-k) filtering, time-variant whitening, and automatic gain control. To address significant near-surface velocity heterogeneity, first-break picks were iteratively refined to generate a refractor model for robust refraction and surface-consistent residual statics. These first breaks were subsequently used to derive turning-ray velocity models for time-to-depth conversion. After evaluating multiple depth imaging techniques, Shallenberger et al. applied Kirchhoff depth migration, demonstrating that it best resolved complex structural geometries, collapsed diffracted energy, and clarified steep dips within the internal stratigraphy (Shallenberger et al., 2026).

The finalized, depth-migrated 2D sections were then exported to Petrel for 3D structural modeling. Shallenberger et al. integrated the orthogonal transects to map coherent subsurface reflection horizons, generating a geographically accurate 3D depth-structure model of the subsurface. This spatial modeling successfully captured regional basin geometries and

complex, fault-like localized deformations. However, Shallenberger et al. note that a low-fold zone at the intersection of the RCR-14 and RCR-22 survey lines resulted in diminished subsurface coverage, requiring careful interpretation of the modeled features in that specific region.

2.3. Preliminary Modeling

The three depth-converted seismic sections were interpreted in Petrel (Petrel, 2023) by mapping lateral continuity and reflection polarity. Because each seismic section was completed as a 2D section, seismic sections RCR-14, RCR-22, and HWY 131 had to be modeled together in a 3D window to mimic a 3D seismic survey (Fig. 2). Polarity standards

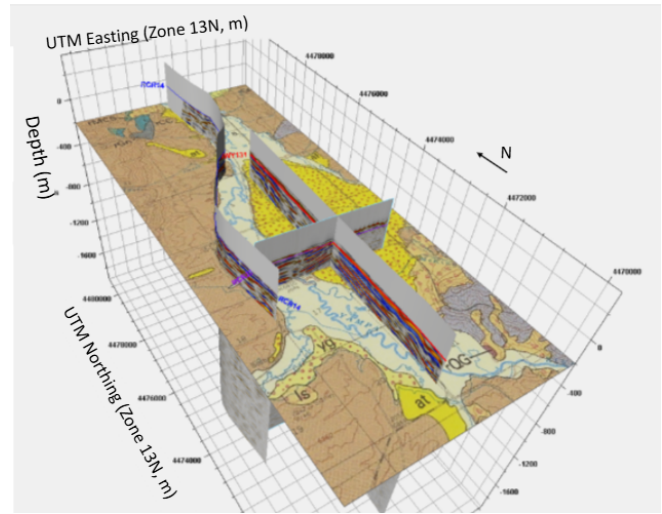


Fig. 2. 3D map of each seismic section

followed SEG convention (*Polarity Standard*, 2026), where a sharp impedance increase with depth produces a normal polarity reflection (peak) and a decrease produces a reverse polarity reflection (trough), assuming zero-phase wavelets. The basement contact was mapped first, as the impedance contrast between overlying sedimentary units and Precambrian crystalline basement produces a characteristically strong normal polarity reflection. This basement pick served as the anchor for all subsequent horizon interpretation, with shallower horizons traced along continuous normal polarity peaks and terminated where polarity reversals or abrupt reflection changes indicated stratigraphic boundaries. From these horizon picks, combined with the velocity and depth models from Shallenberger et al. (2026), Petrel was used to generate a 3D model of the basement surface and overlying stratigraphic layers across the study area.

Discontinuities in reflection continuity, including offsets, truncations, and chaotic reflection packages, were interpreted as indicators of structural deformation such as faulting. These disruptions informed horizon routing, where lateral continuity alone was insufficient to constrain layer geometry.

3. RESULTS

From interpretation of each survey line, we constructed geologic cross-sections emphasizing fault geometry and stratigraphic features that are obscured in the raw seismic amplitude data. Additional interpolated 2D cross-sections were generated within the survey region to extend interpretation between lines.

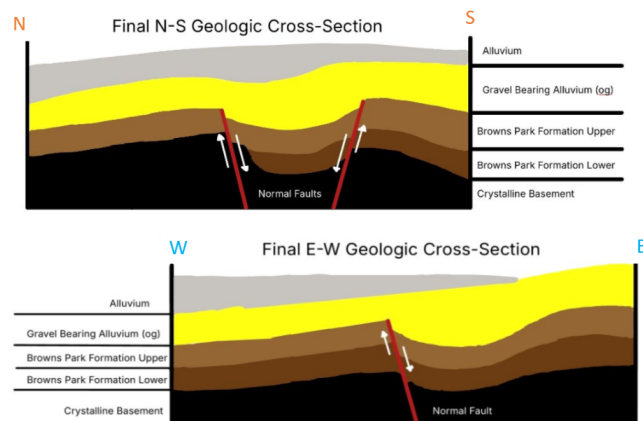


Fig. 3. Final interpreted geologic cross-section

Interpretation of the three depth-migrated seismic sections yielded five distinct stratigraphic layers, shown in the final geologic cross-sections (Fig. 3). From bottom to top, these are: Precambrian crystalline basement, a lower Browns Park unit (BP2), an upper Browns Park unit (BP1), a gravel-bearing glacial alluvium, and a surficial alluvial unit. The basement surface produces a strong, laterally continuous normal polarity reflection and exhibits southward dip along the N-S section. The two Browns Park packages are distinguished by separate continuous horizon picks at different depths. Both exhibit normal polarity reflections consistent with distinct depositional surfaces.

Normal faults are visible in both the N-S and E-W cross-sections as abrupt offsets in the BP1 and BP2 reflectors (Fig. 3). Along the E-W section, a fault on the eastern end displaces the Browns Park horizons with the hanging wall down to the east. Along the N-S section, fault displacement is expressed as southward thickening of the gravel-bearing alluvial unit into the accommodation space created by fault offset. The overlying alluvial unit remains relatively undisturbed across both sections. These horizon picks were connected across all three 2D sections and interpolated in Petrel to generate a 3D structural model of the subsurface (Fig. 4), capturing basin geometry and fault plane orientations across the full survey area.

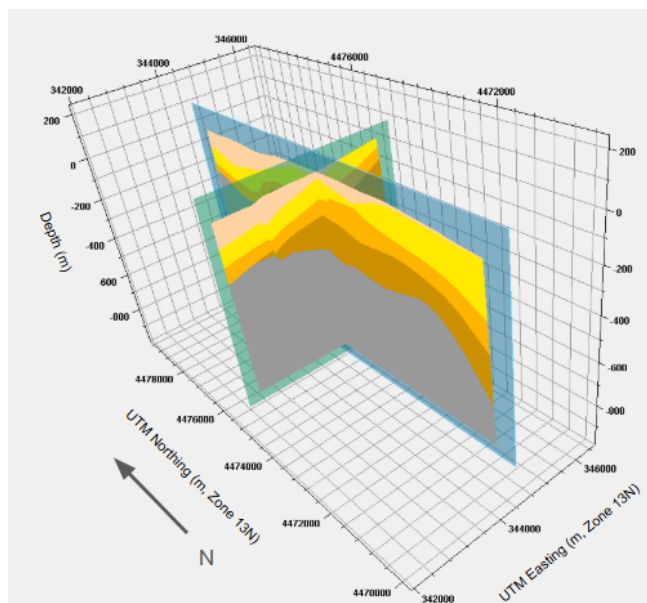


Fig. 4. Cross-section of 3D model

4. DISCUSSION

We interpret normal faulting along RCR-22 and between RCR-14 and CO State Highway 131 based on the interpreted cross-sections. Fig. 3 shows the five distinct stratigraphic layers. The bottom layer is interpreted as crystalline basement rock, identified by its characteristically high seismic velocity and strong reflectivity. Overlying the basement are two basin packages, an upper Browns Park unit (BP1) and a lower Browns Park unit (BP2), both assigned to the Browns Park Formation but representing different depositional ages. Following deposition of BP1 and BP2, we interpret the presence of a normal fault. The timing of this faulting is constrained using the principle of original horizontality (National Park Service, n.d.), which states that sedimentary layers are deposited horizontally and parallel to the Earth's surface; any deviation from this original geometry therefore reflects post-depositional deformation. The offset observed in BP1 and BP2 indicates that faulting post-dates their deposition, while the relatively undisturbed geometry of the overlying glacial and alluvial units suggests that significant fault displacement had ceased prior to their deposition.

Overlying the faulted sequence, we interpret a glacial gravel unit deposited during the last glacial maximum, approximately 18,000–12,000 years ago. The southward thickening of this unit is consistent with glacial outwash prograding into the accommodation space created by fault displacement. Following deglaciation, the Yampa River likely established its course within the glacially carved valley, with the uppermost alluvial unit representing continued fluvial deposition.

Although additional seismic coverage is needed to confirm this interpretation, the trajectory of the Yampa River pro-

vides supporting evidence. Rivers are known to incise along fault-weakened rock zones, and the linear trajectory of the Yampa River parallel to RCR-14 suggests structural control on its drainage path (Vita-Finzi, 2012; Holbrook & Schumm, 1999). Taken together, the reflection geometry, stratigraphic offsets, and river trajectory lead us to interpret a fault near or along RCR-14.

4.1. Interpretation Limitations

Interpretation confidence was limited by two primary factors. First, time constraints restricted our analysis to a single iteration of the Petrel horizon interpretation. Given additional time, multiple independent interpretations of the same dataset could be generated and compared, providing a more robust assessment of interpretation uncertainty. Second, borehole data, where available, can be incorporated into Petrel to independently constrain horizon depths and improve stratigraphic correlation. No borehole data exist within our survey area, precluding this additional constraint and leaving horizon picks reliant solely on seismic reflection character and lateral continuity.

5. FUTURE WORK

To create a more refined and comprehensive interpretation, we recommend running more east-west seismic survey lines to further constrain the suspected fault plane that the Yampa River follows, as well as an extension of RCR-14 to verify features seen on the south end of HWY 131. This would allow us to observe whether the structures remain consistent or vary across this area of the basin. We also know that several other seismic lines have been acquired in previous field camps, with 2 lines being surveyed west of RCR-14, acquired in 2025, and two lines being surveyed northwest of the city of Steamboat Springs, acquired in 2024. Time was limited to incorporate these surveys into our seismic interpretations beyond a brief comparison, but they could be used to characterize the geology of a much wider area, and may enhance our existing interpretations. Future surveys near known borehole locations would provide independent stratigraphic constraints and enable correlation of seismic horizons across a wider region.

6. CONCLUSIONS

Reflection seismic interpretation of the Yampa River Basin south of Steamboat Springs reveals a five-layer stratigraphic model comprising Precambrian crystalline basement, two distinct depositional packages of the Browns Park Formation, a glacial gravel unit, and a surficial alluvial unit. Normal faulting post-dating Browns Park deposition and pre-dating glaciation is interpreted along RCR-22 and between RCR-14

and HWY 131, with an additional fault inferred near RCR-14 from stratigraphic offsets and the structurally controlled trajectory of the Yampa River. These fault structures have not previously been mapped in this area and represent a new constraint on the subsurface architecture of the basin.

These findings have broader implications for understanding the Steamboat Springs region. Fault-controlled fluid pathways are a well-established mechanism for hydrothermal spring emergence, and the structures identified may play a role in controlling the distribution of the area's hot springs. Continued geophysical investigation, particularly additional east-west survey lines and integration with previous field camp datasets, will be essential to further constrain fault geometry and refine our understanding of basin evolution.

7. AUTHOR CONTRIBUTION STATEMENT

A.L. and L.H. constructed the 2D and 3D structural models in Petrel and co-authored the Methods section. B.K. and Z.S. contributed to the Methods section. A.L. authored the Results section and co-produced the geologic cross-sections with A.S. A.S. led geologic interpretations, co-produced the geologic cross-sections, and authored the Discussion. L.H. and A.S. co-authored the Introduction and Abstract. B.K. and Z.S. provided critical review and editorial revisions across the manuscript.

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